MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF MATHEMATICS

IV SEMESTER B.TECH. (MATH & COMPUTING)

MID SEMESTER MAKE UP EXAMINATION

VECTOR ANALYSIS AND COMPLEX VARIABLES–MAT 2238

Date of Examination: 11 April 2025 Duration: 90 mins MAX.MARKS: 30

Part B: DESCRIPTIVE

Answer all the questions.

Q. No	Question	Marks
1	For $0 < \theta < 2\pi$, derive the identity $$1 + \cos\theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin[(2n+1)\theta/2]}{2\sin\theta/2}$$ Hint: Use the identity $$1 + z + z^2 + \cdots + z^n = \frac{1 - z^{n+1}}{1 - z}, z \neq 1$$	**5**
2	Let $ln(a)$ denote the natural logarithm to the base e for a positive real a. Also, let $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$. Compute the gradient of $\phi(x,y,z) = ln\, r$, where, $r = \parallel \vec{r} \parallel$ at (2,0,0).	**4**
3	Find an equation for the tangent plane to the surface $$x^2yz - 4xyz^2 = -6$$ at the point (1,2,1).	**3**
4	Find the angle between the surfaces $z = x^2 + y^2$ and $z = (x - \frac{1}{\sqrt{6}})^2 + (y - \frac{1}{\sqrt{6}})^2$ at the point $(\frac{\sqrt{6}}{12}, \frac{\sqrt{6}}{12}, \frac{1}{12})$.	**3**

5	Let $\vec{v}(x, y, z)$ be a continuously differentiable vector field that is also conservative. Then, determine $curl(\vec{v})$.	**3**
6	Find all the complex cube roots of $-8i$ in rectangular coordinates.	**4**
7	Determine all the points in the complex plane where the function $f(z) = Re(z)$ is differentiable. Justify your answer.	**3**